

02477 – Bayesian Machine Learning: Lecture 9

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Outline

- 1 Markov chain Monte Carlo
- 2 Advanced sampling methods
 - Hamiltonian dynamics
 - Hamiltonian Monte Carlo
 - The MALA and ULA algorithms
- 3 Convergence diagnostics

Markov chain Monte Carlo

Bayesian inference and probabilistic modelling

■ Bayesian supervised learning in general:

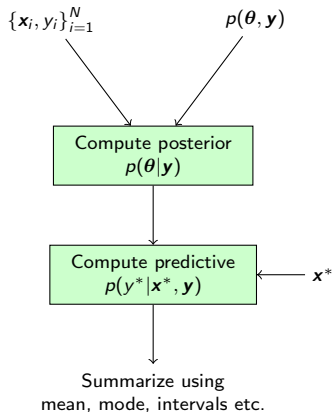
1. Joint model of data $\mathbf{y}$ and parameters θ .
2. Summarize knowledge of θ given data $\mathbf{y}$.
3. Compute posterior predictive distribution.

■ Goal: separate modelling from inference:

1. Build models reflecting domain knowledge.
2. Push “inference button” and get results.

■ Inference methods:

1. Laplace approximations.
2. Markov chain Monte Carlo.
3. Variational approximations.



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- We showed last time that ...

1. the Monte Carlo estimator $\hat{f}$ is *unbiased*
2. the variance of $\hat{f}$ decreases with $1/S$ when the samples are *i.i.d.*

A zoo of sampling-based methods

■ Simple sampling methods

1. Rejection sampling
2. Ancestral sampling
3. Importance sampling
4. Transformation methods
5. Inverse transform sampling
6. ...

■ MCMC methods

1. Metropolis-Hastings
2. Gibbs Sampling
3. Slice sampling
4. Hamiltonian Monte Carlo
5. ...

MCMC using the Metropolis-Hastings algorithm

- We can use the MH to generate samples from a distribution of interest $p_t(\theta)$.

The Metropolis-Hastings algorithm

- Start from some initial value θ^0 (e.g., a sample from the prior).
- Repeat for $k = 1$ to K :
 1. Given last value θ^{k-1} , generate *candidate sample* using proposal distribution

$$\theta^* \sim q(\theta^* | \theta^{k-1}).$$

2. Compute *acceptance probability* A_k as follows

$$A_k = \min \left(1, \frac{p_t(\theta^*)q(\theta^{k-1} | \theta^*)}{p_t(\theta^{k-1})q(\theta^* | \theta^{k-1})} \right).$$

3. *Accept* θ^* with probability A_k

$$\theta^k = \begin{cases} \theta^* & \text{if } u_k < A_k \\ \theta^{k-1} & \text{otherwise} \end{cases}, \quad \text{for } u_k \sim \mathcal{U}(0, 1)$$

- What do we need in order to implement MH for a given model?
- High flexibility, but the price we pay is correlation in generated samples

Markov chain Monte Carlo theory I

- Metropolis-Hastings defines a chain of samples $\theta^0, \theta^1, \theta^2, \dots$ with a *Markov property*

$$p(\theta^{k+1} | \theta^1, \theta^2, \dots, \theta^k) = p(\theta^{k+1} | \theta^k)$$

- The *transition kernel* tells us how to iterate the chain

$$T(\theta^{k+1} | \theta^k) \equiv p(\theta^{k+1} | \theta^k)$$

- The distribution of θ^{k+1} is given by *sum rule*

$$p(\theta^{k+1}) = \int T(\theta^{k+1} | \theta^k) p(\theta^k) d\theta^k$$

- A distribution $p^*(\theta)$ is said to be *invariant* or *stationary* wrt. the Markov chain if each step does not change the distribution

$$p^*(\theta) = \int T(\theta | \theta') p^*(\theta') d\theta'$$

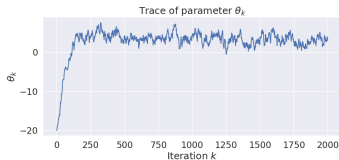
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Transition kernel for Metropolis-Hastings

- Recall the acceptance probability for Metropolis-Hastings

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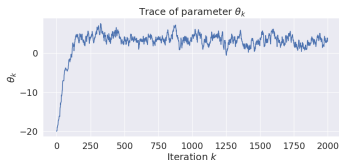
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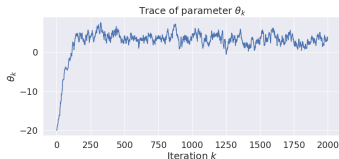
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- The big picture

- We set θ^0 to some initial value (*initialization*)
- We iterate $\theta^{k+1}|\theta^k \sim T(\theta^{k+1}|\theta^k)$ (*warm-up phase*)
- Eventually the distribution of θ^k will converge to the stationary distribution p^* (*sampling phase*)



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- A chain that satisfies (A1)–(A3) is said to be *ergodic* and ensures $p(\theta^k) \rightarrow p^*(\theta)$.

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- For example, all (non-degenerate) Gaussian proposals lead to detailed balance, but convergence time may vary drastically depending on the proposal variance.
- More theory and details: Monte Carlo Statistical Methods by Robert and Casella.



Pros and cons of Metropolis-Hastings

Pros

- Strong mathematical guarantees: If we sample long enough, the iterates θ^k will converge to the exact target distribution

$$p(\theta^k) \rightarrow p^*(\theta) \quad \text{for} \quad k \rightarrow \infty$$

- Easy to implement.
- Easy to prototype and evaluate different models.

Cons

- May have to sample “infinitely” long for difficult distributions.
- Acceptance ratio can be low.
- Slow for large datasets.
- Proposal distribution may require tuning.

Questions: True or false?

Quiz via DTU Learn:

Lecture 9: Metropolis-Hastings (12 questions)

Check you knowledge

Advanced sampling methods

Designing more efficient algorithms

- We have now studied the Metropolis-Hastings algorithm with Gaussian proposals:

$$q(\boldsymbol{\theta}^* | \boldsymbol{\theta}^k) = \mathcal{N}(\boldsymbol{\theta}^* | \boldsymbol{\theta}^k, \tau^2 \mathbf{I}) \quad \Longleftrightarrow \quad \boldsymbol{\theta}^* = \boldsymbol{\theta}^k + \tau \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

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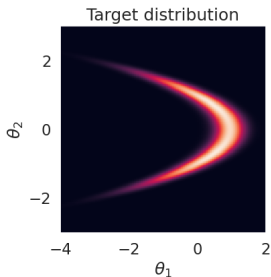
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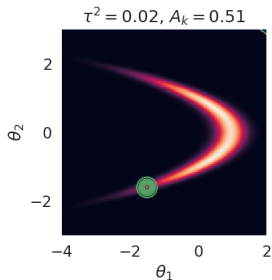


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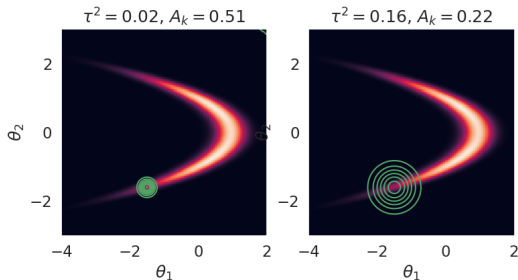


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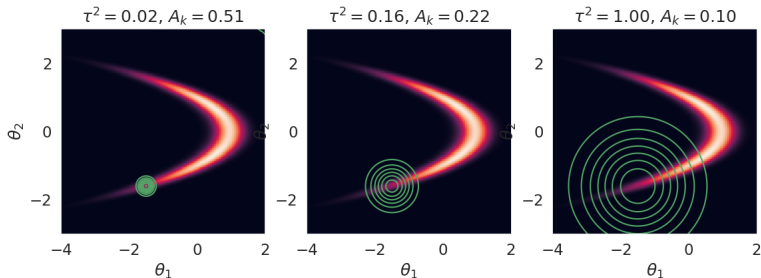


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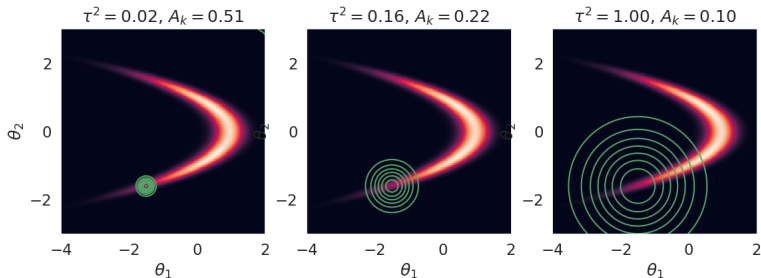


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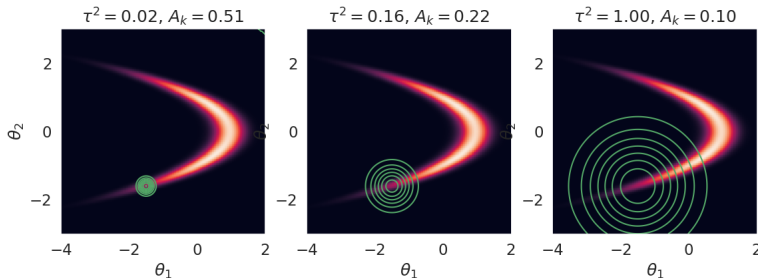
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- Large proposal variance leads to proposals, where target density $p_t(\theta)$ is very low
- Structure of target distribution forces us to use small proposal variance, which leads to slow exploration. This problem becomes even *worse in high-dimensional* spaces.

Hamiltonian Monte Carlo

- The *Hamiltonian Monte Carlo* (HMC) is the state-of-the-art MCMC algorithm
 - 1987: First proposed by team working in quantum physics
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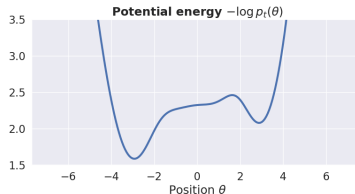
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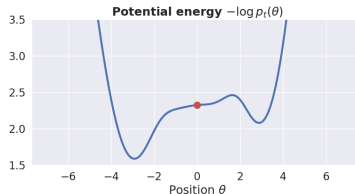
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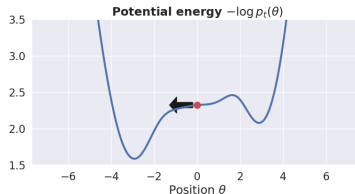
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Hamiltonian Monte Carlo Animation

Show animation for intuition

<https://chi-feng.github.io/mcmc-demo/app.html>

Advanced sampling methods: Hamiltonian dynamics

Hamiltonian Monte Carlo: Definitions and terminology

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- Let $K(\eta) = -\log p(\nu) + \text{const}$ denote the *kinetic energy*
- Define the *Hamiltonian function* $H(\theta, \nu)$ as the *total energy*

$$H(\theta, \nu) = E(\theta) + K(\nu) + \text{const}$$

Hamiltonian Monte Carlo: Definitions and terminology

- Let $p_t(\theta)$ be the target distribution for parameters of interest $\theta \in \mathbb{R}^D$. We refer to θ as the *position* of some imaginative particle.
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- Let $E(\theta) = -\log p_t(\theta) + \text{const}$ denote the *potential energy*.
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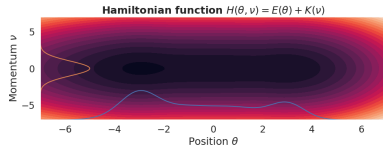
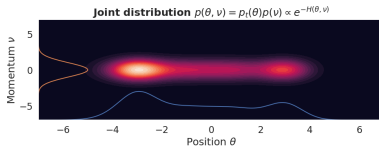
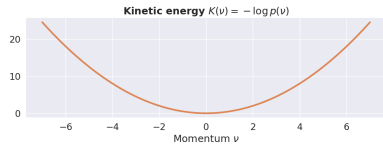
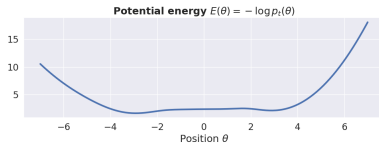
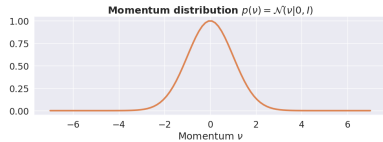
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- Assume the joint distribution over *phase space* (θ, ν) is given by

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Hamiltonian Monte Carlo: 1D Example

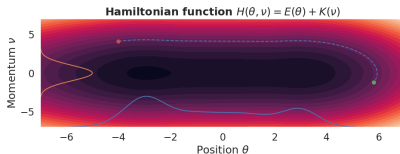


Hamilton's equations

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- *Hamilton's equations* describes how a particle with position θ and momentum ν moves over time

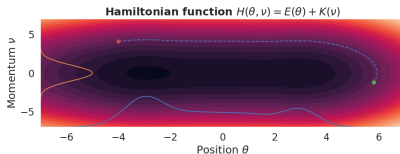
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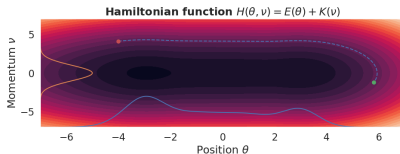
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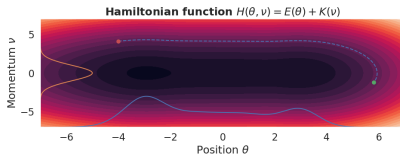
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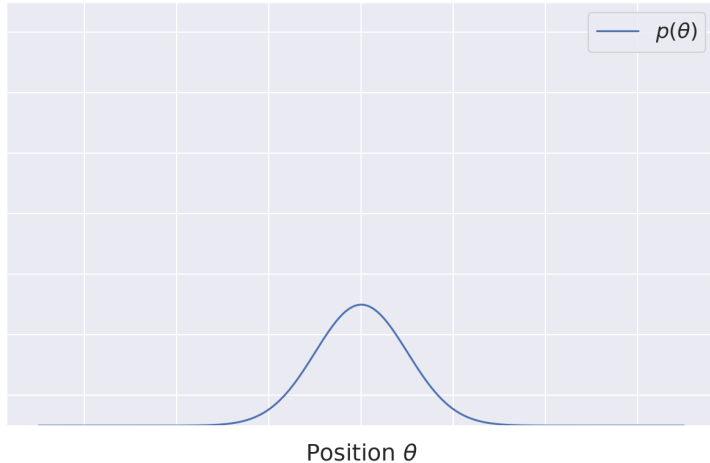


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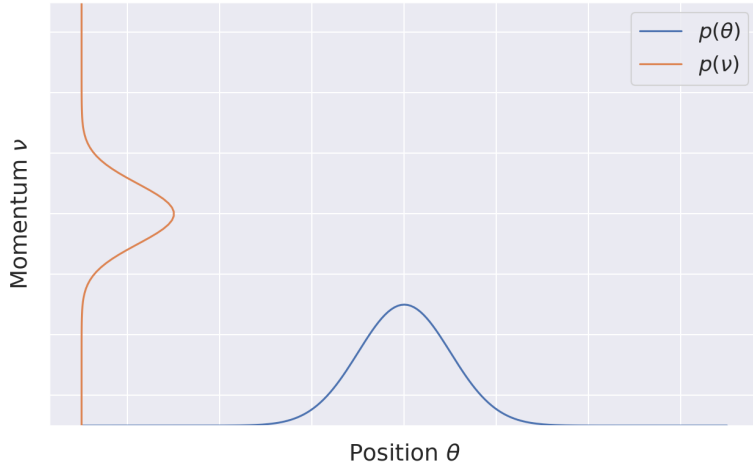
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- We can use Hamiltonian dynamics to *propose large steps* in θ while avoiding random walk behavior!

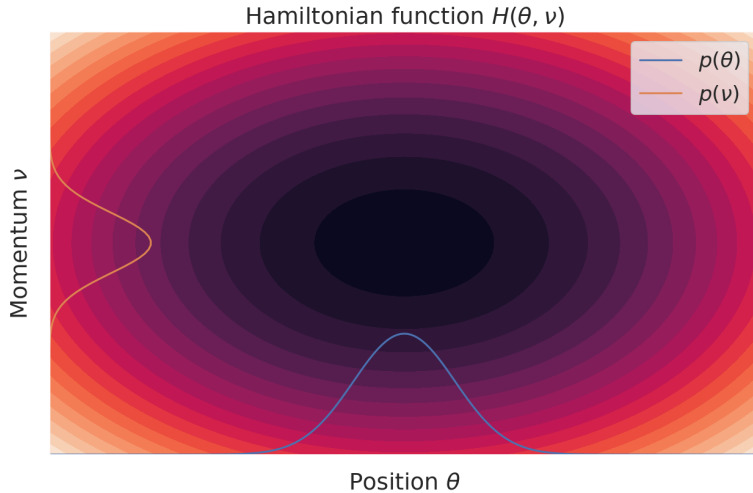
Hamiltonian Monte Carlo: Examples



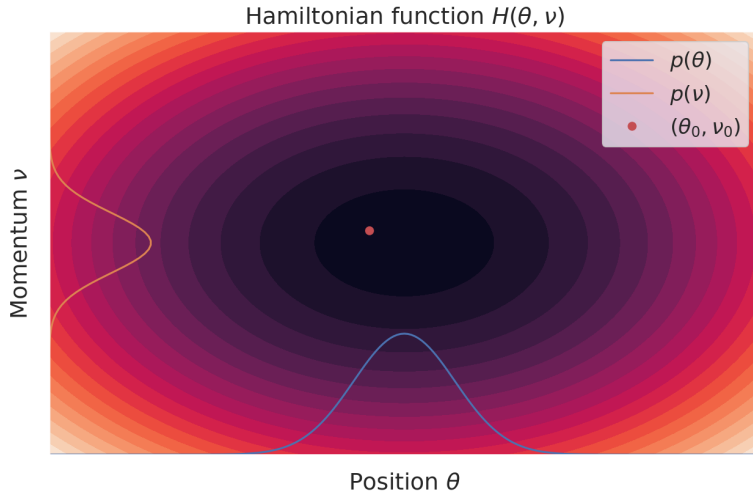
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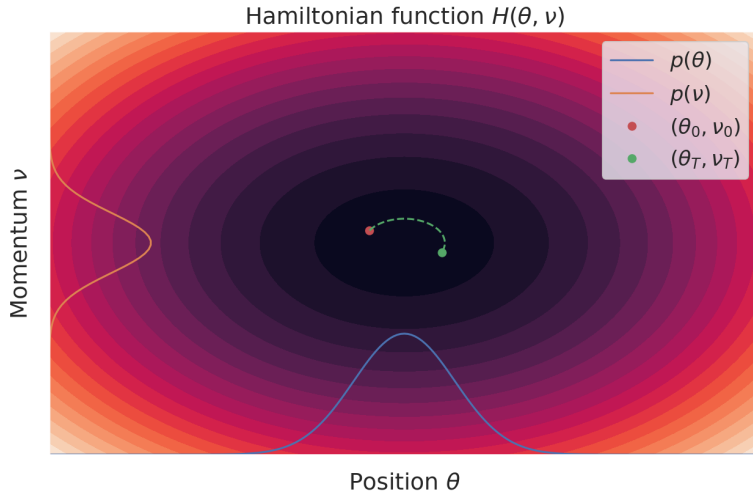
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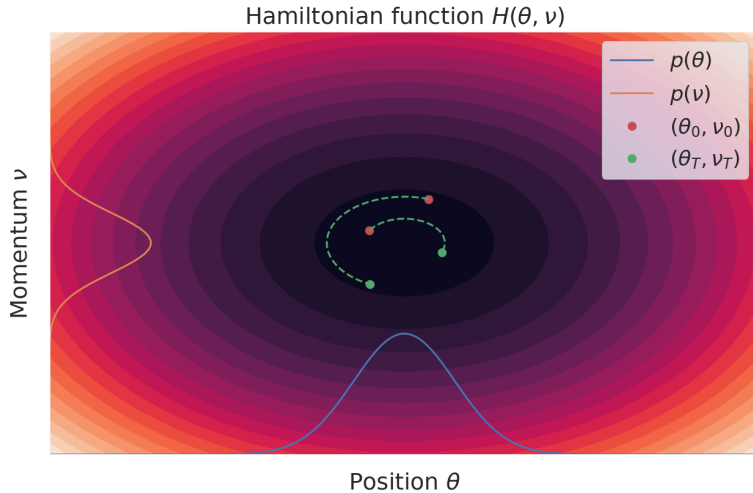
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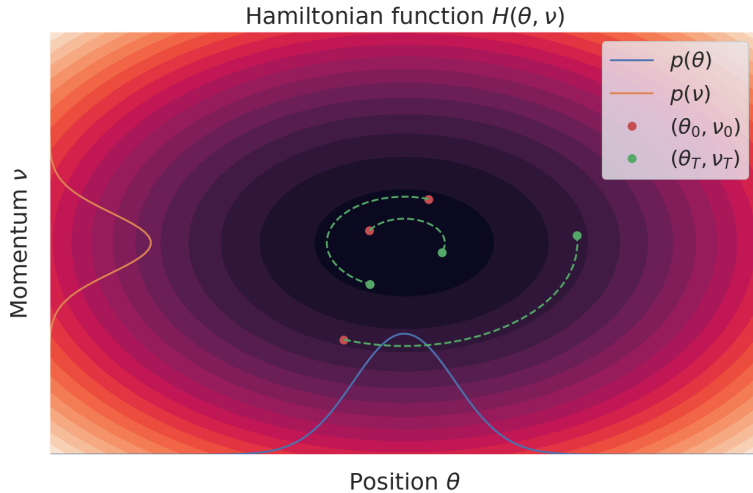
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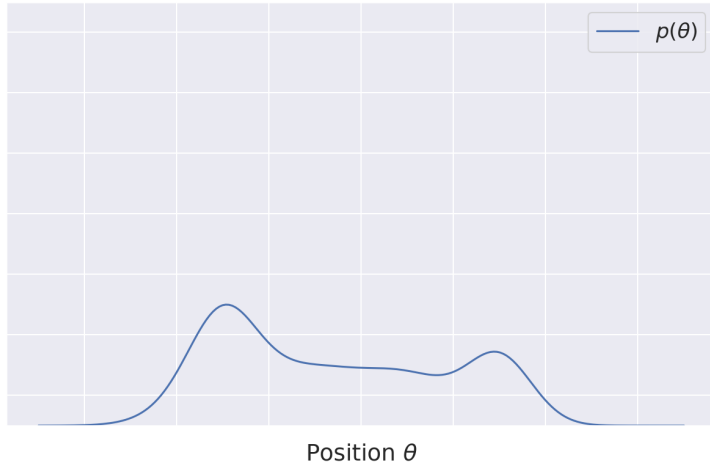
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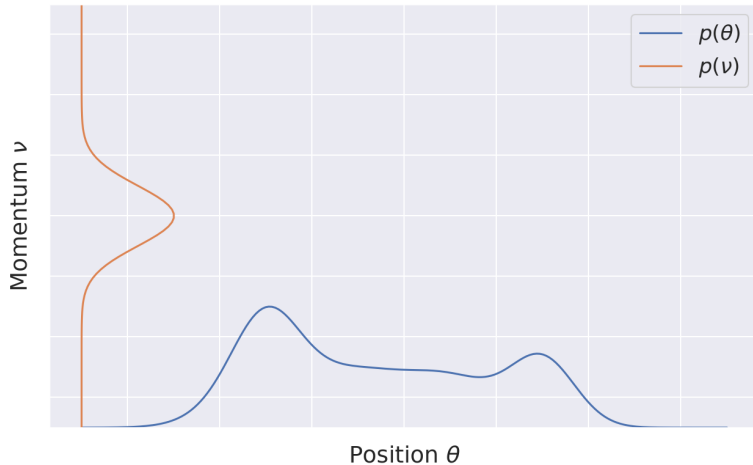
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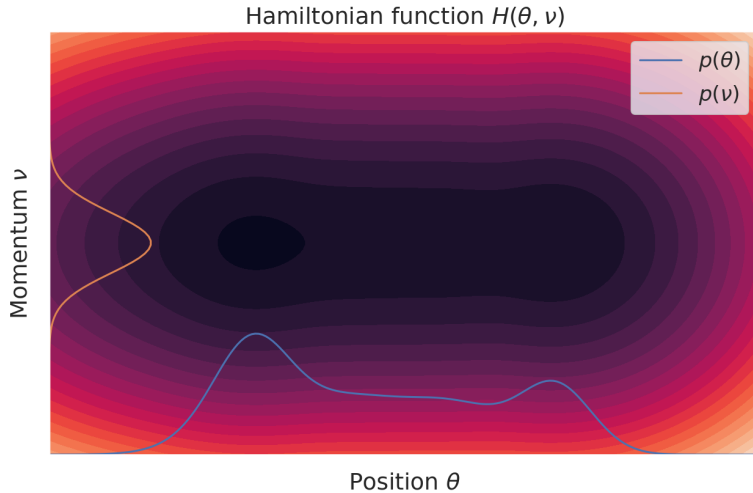
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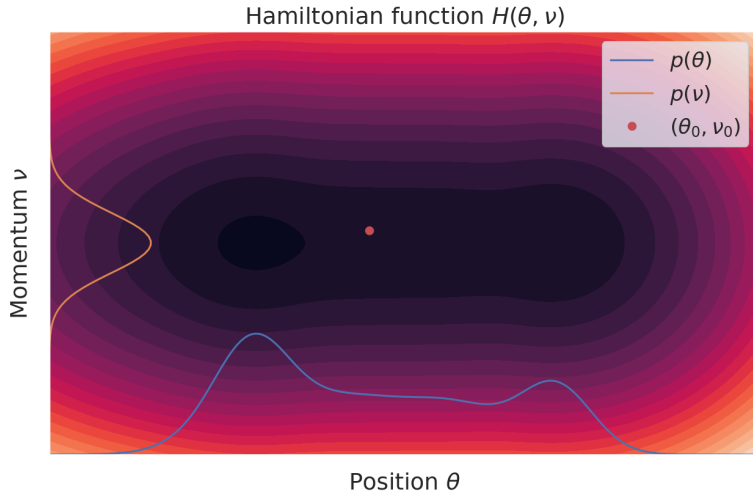
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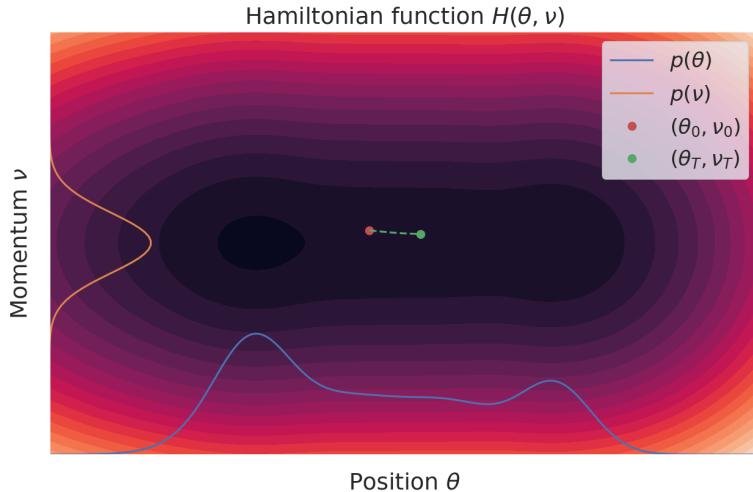
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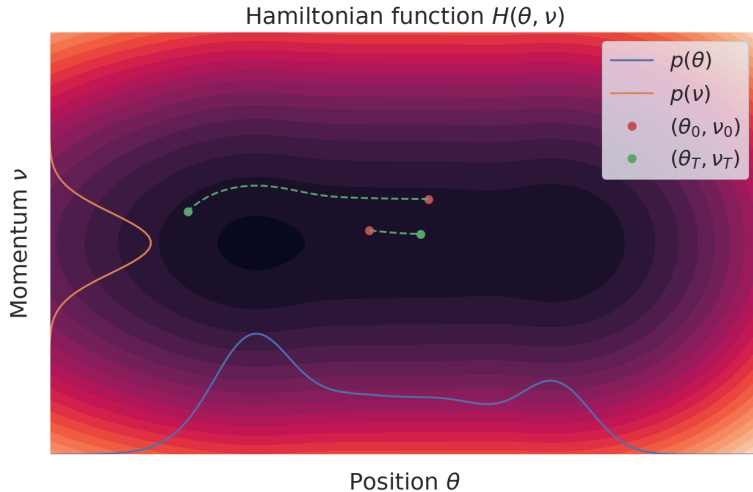
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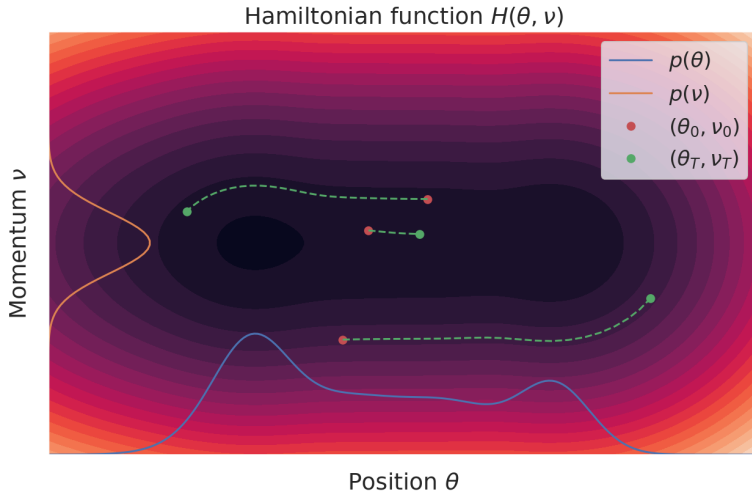
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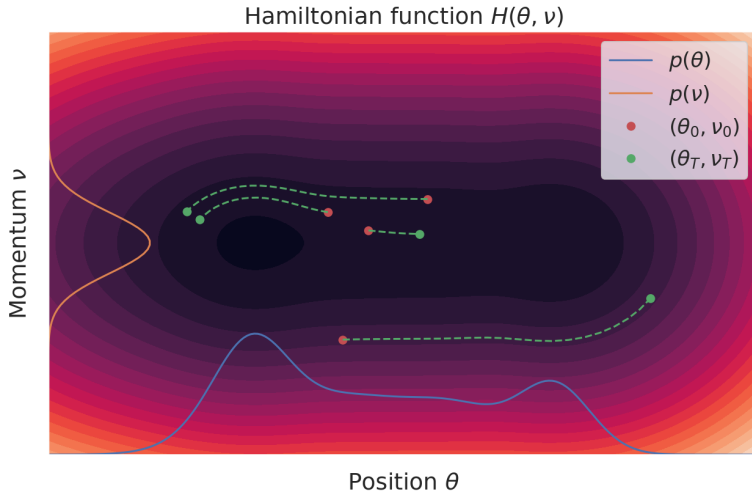
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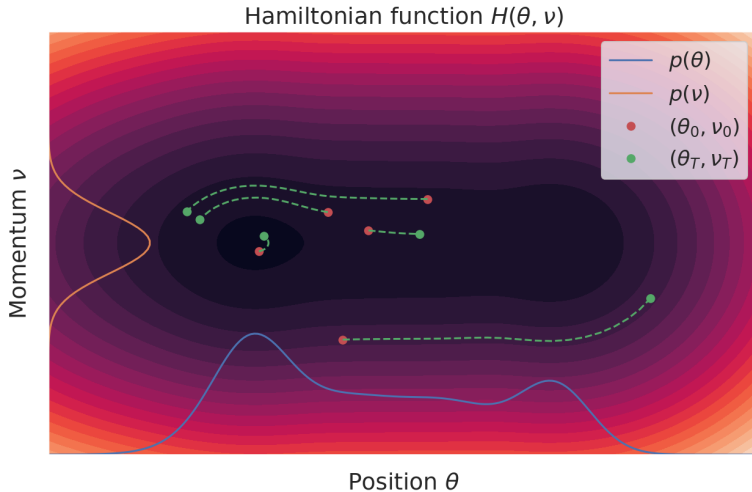
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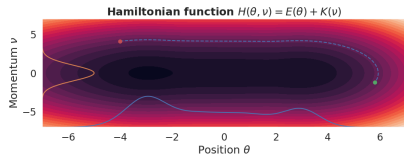
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Integrating Hamilton's equation: Euler's method

- We *discretize* and solve *Hamilton's equation* numerically

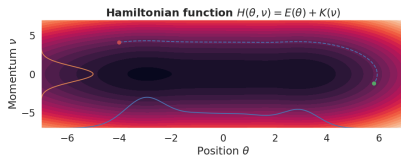
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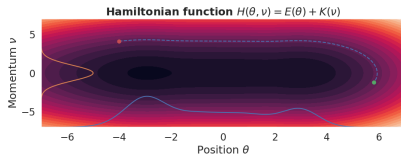


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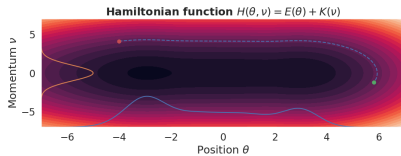
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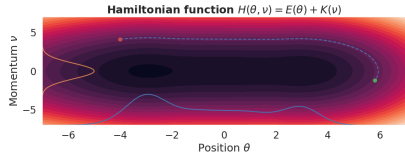
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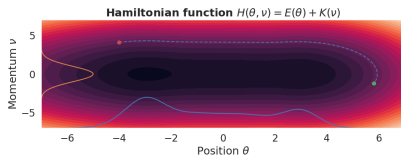
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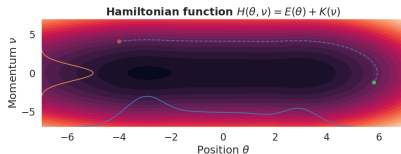
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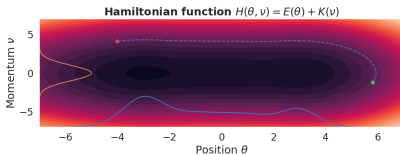
$$\begin{aligned}\theta_{k+1} &\approx \theta_k + \eta \nu_k \\ \nu_{k+1} &\approx \nu_k - \eta \nabla_{\theta} E(\theta_k)\end{aligned}$$

- Because of the *approximation error* introduced by Euler's method, we *cannot* turn this into a valid and symmetric proposal distribution

Integrating Hamilton's equation: The leapfrog method

- We use the *leapfrog method* instead because it is/has:

1. Reversible
2. Volume-preserving ($|\det J| = 1$)
3. Better approximation error



- Given a *step-size* $\eta > 0$ and a state (θ_k, ν_k) at time $t = k\eta$, we can approximate $(\theta_{k+1}, \nu_{k+1})$ using a *leapfrog integrator*:

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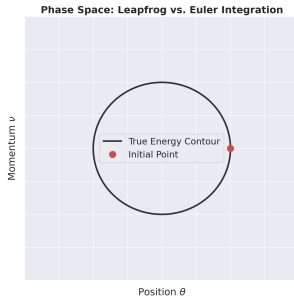
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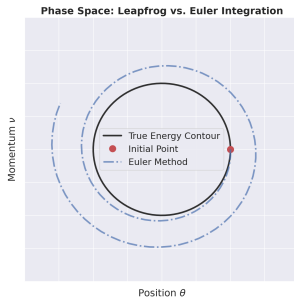
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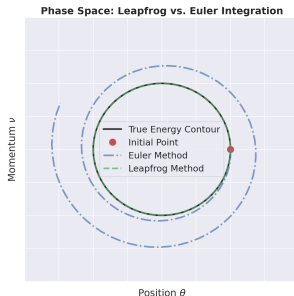
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2. Volume-preserving ($|\det J| = 1$)
3. Better approximation error



- Given a *step-size* $\eta > 0$ and a state (θ_k, ν_k) at time $t = k\eta$, we can approximate $(\theta_{k+1}, \nu_{k+1})$ using a *leapfrog integrator*:

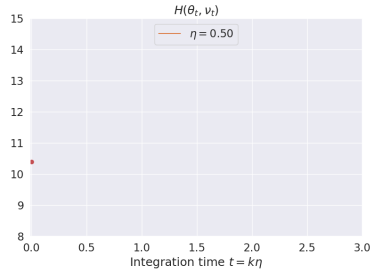
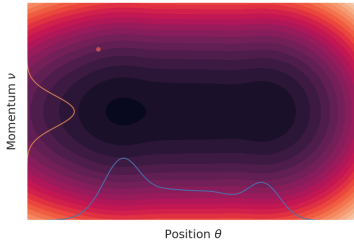
$$\nu_{k+1/2} = \nu_k - \frac{\eta}{2} \nabla_{\theta} E(\theta_k) \quad (\text{Half-step for } \nu)$$

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- The above equations correspond to *1 step* of the leapfrog integrator. We *iterate* these equations for L steps to approximate the state at time $T = L\eta$

Leapfrog integration in action I



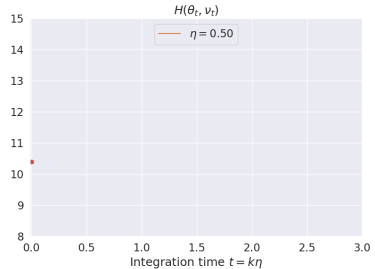
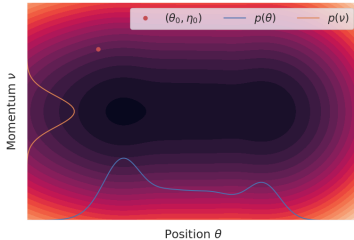
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Leapfrog integration in action I



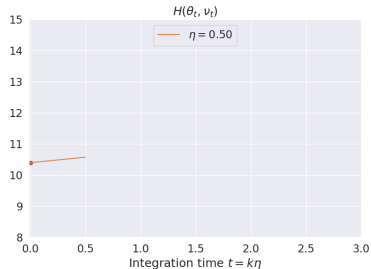
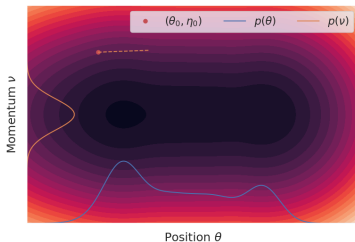
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Leapfrog integration in action I



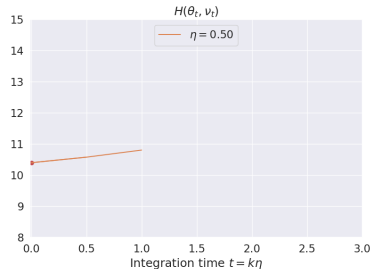
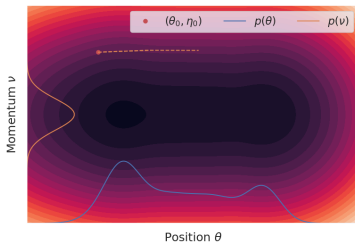
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Leapfrog integration in action I



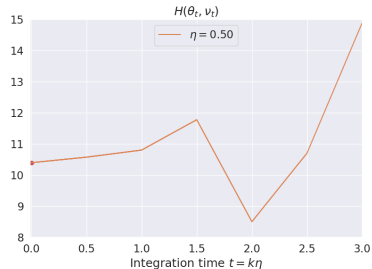
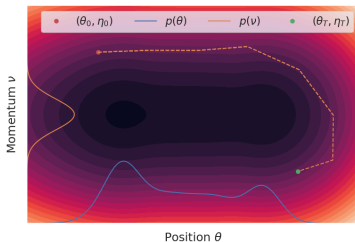
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Leapfrog integration in action I



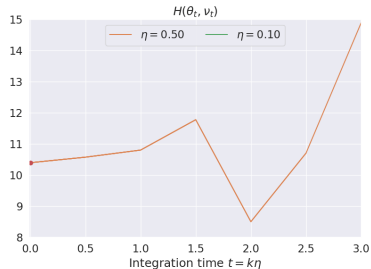
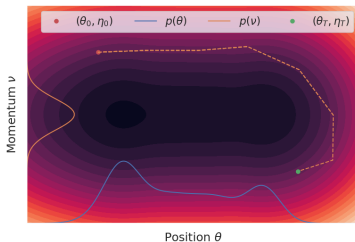
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Leapfrog integration in action I



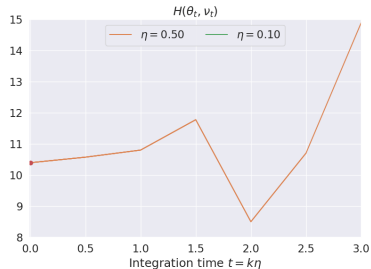
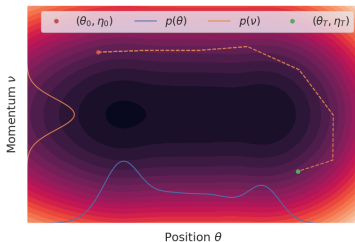
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Leapfrog integration in action I



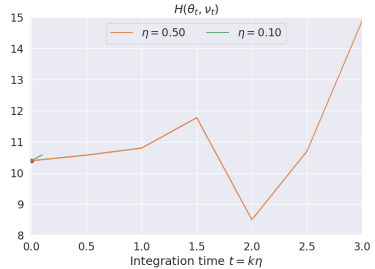
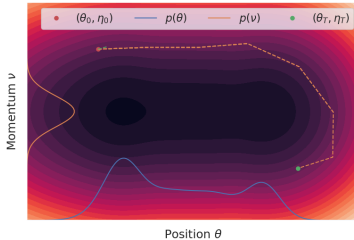
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Leapfrog integration in action I



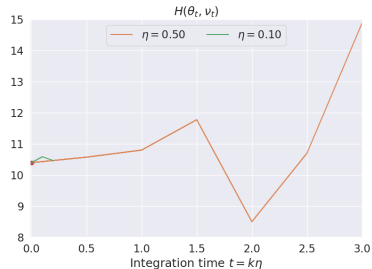
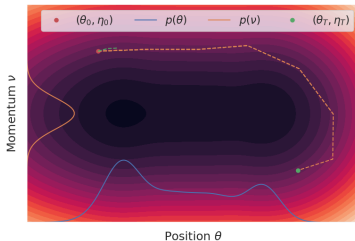
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Leapfrog integration in action I



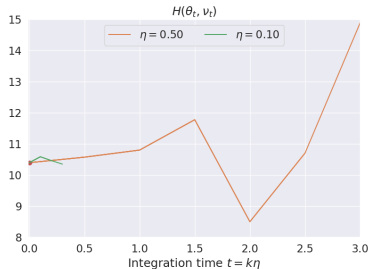
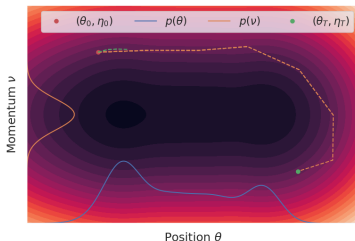
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Leapfrog integration in action I



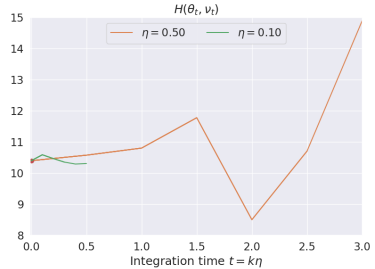
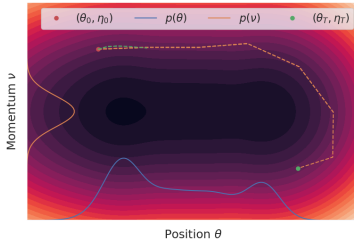
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Leapfrog integration in action I



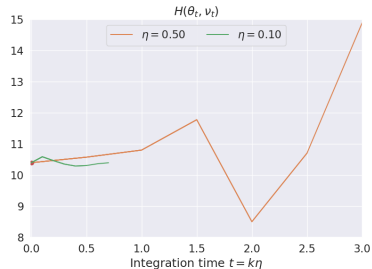
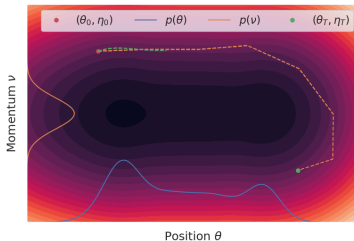
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Leapfrog integration in action I



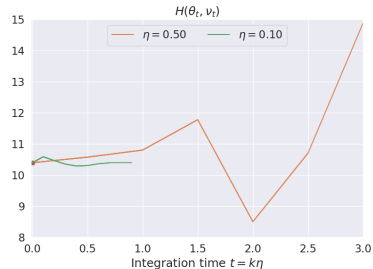
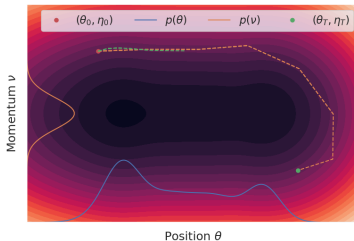
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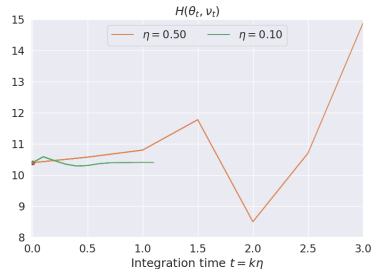
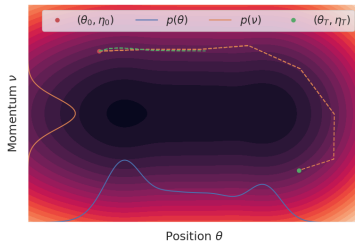
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Leapfrog integration in action I



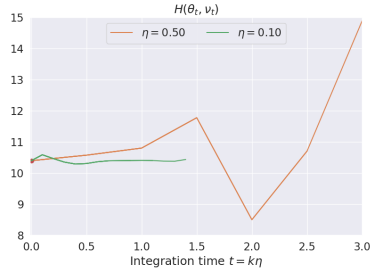
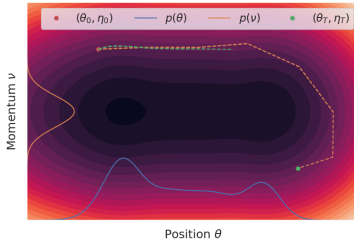
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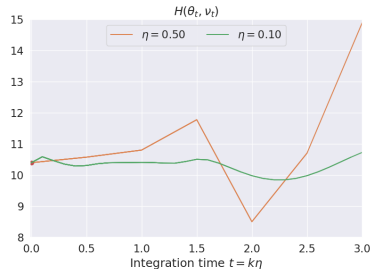
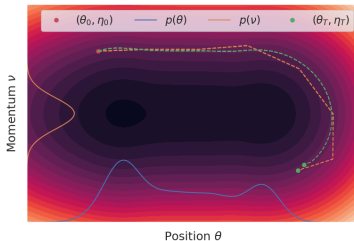
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Leapfrog integration in action I



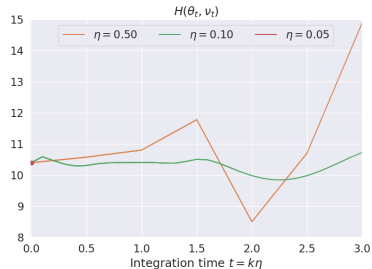
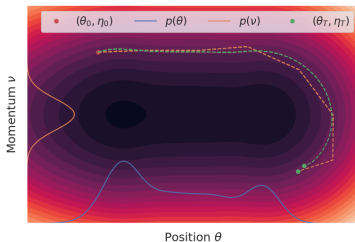
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Leapfrog integration in action I



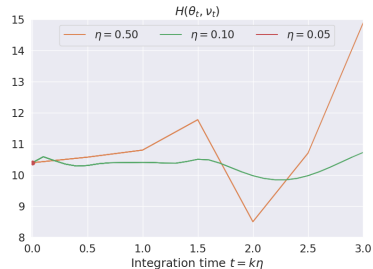
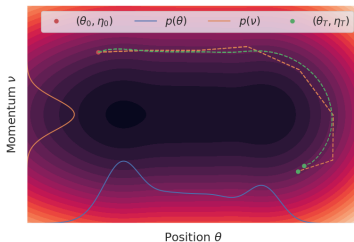
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Leapfrog integration in action I



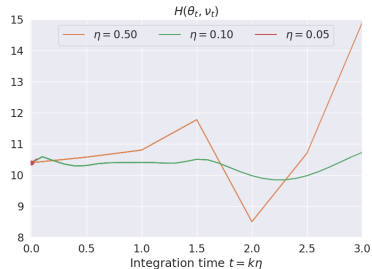
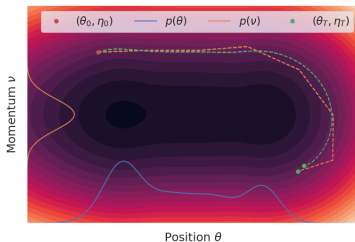
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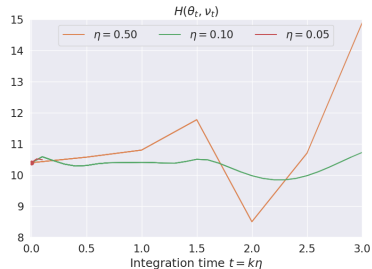
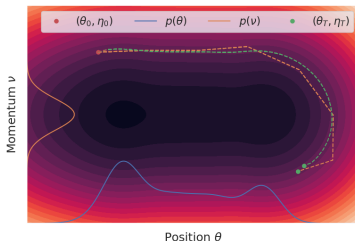
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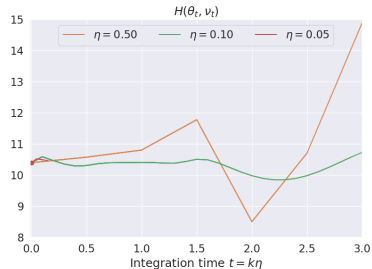
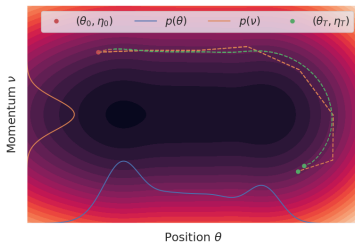
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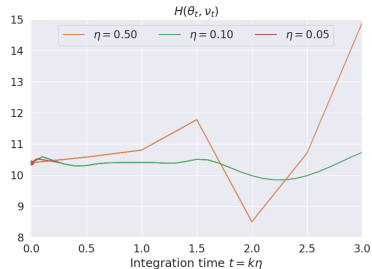
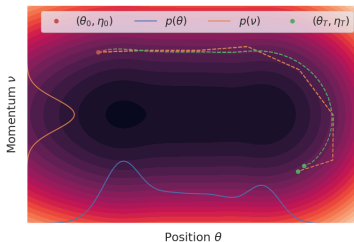
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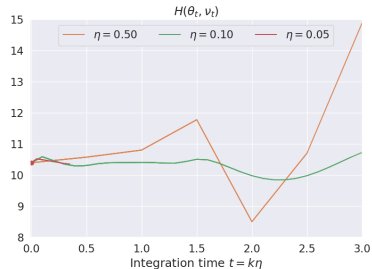
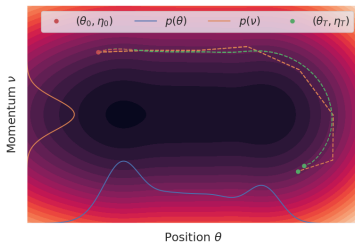
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Leapfrog integration in action I



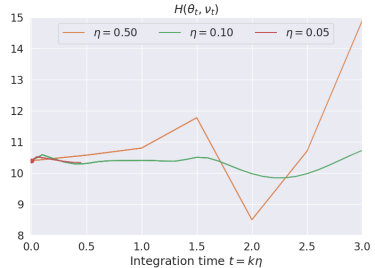
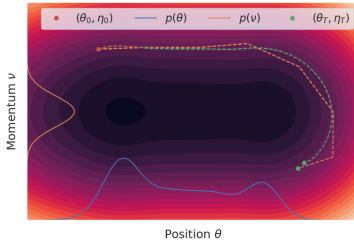
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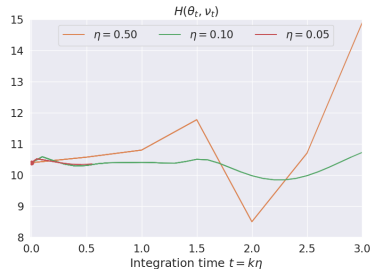
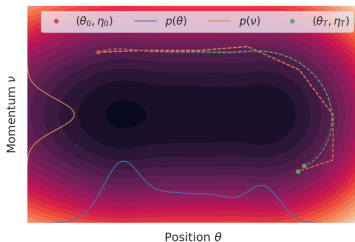
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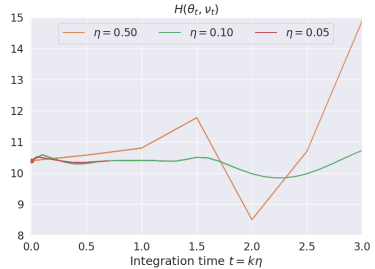
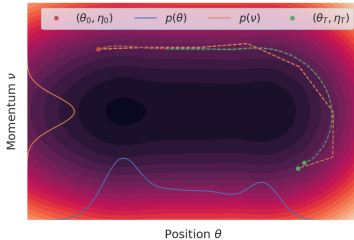
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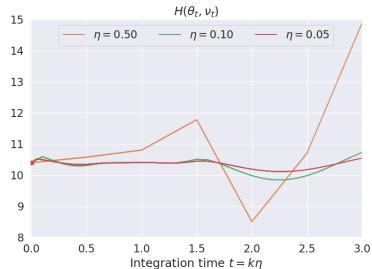
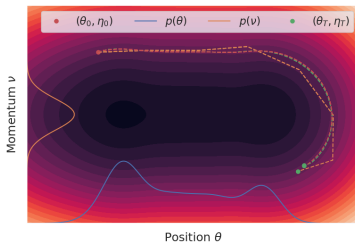
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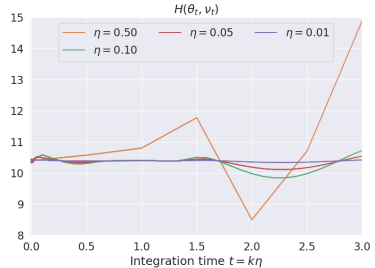
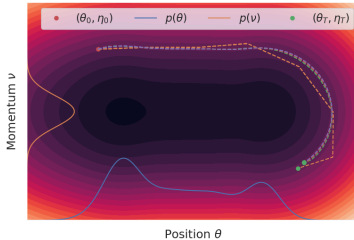
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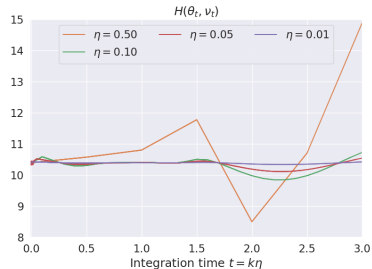
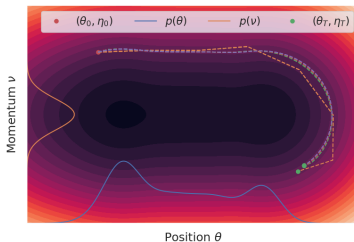
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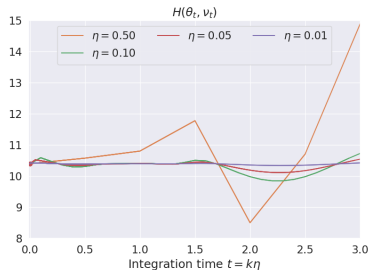
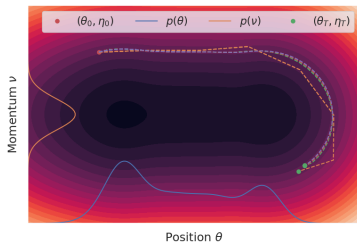
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The leapfrog procedure defines a *deterministic map* $(\theta_k, \nu_k) \mapsto (\theta^*, \nu^*)$, i.e.

$$(\theta^*, \nu^*) = \text{leapfrog}_{\eta, L}(\theta_k, \nu_k)$$

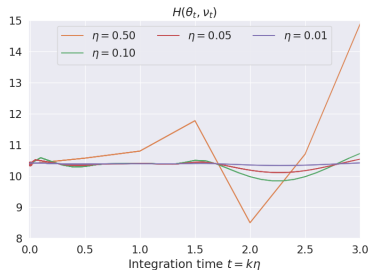
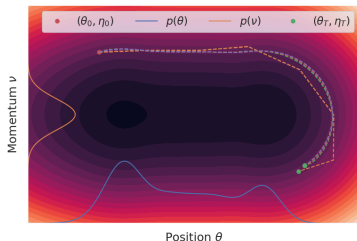
Leapfrog integration in action II



A few observations

- The total integration time is given by $T = L\eta$, where the step-size η determines the accuracy

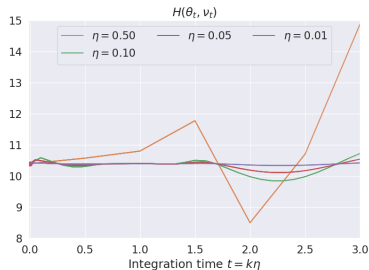
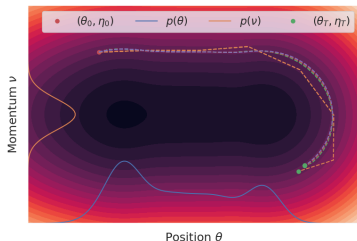
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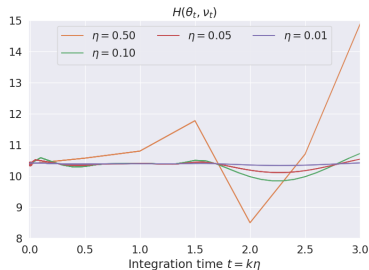
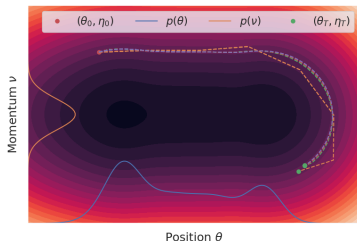
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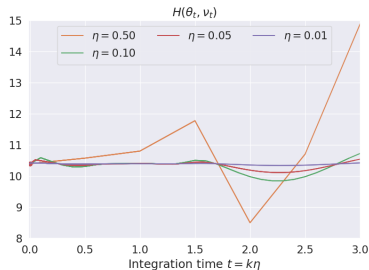
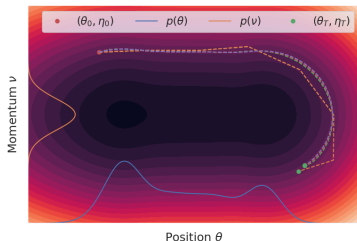
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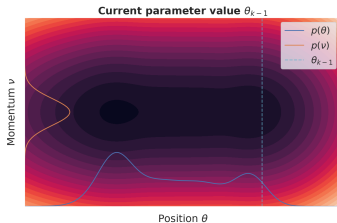


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- In practice, we have robust heuristics to automatically tune both L and η
- The Hamiltonian $H(\theta, \nu)$ is *only approximately preserved in practice*, but this is not a problem as we will see soon

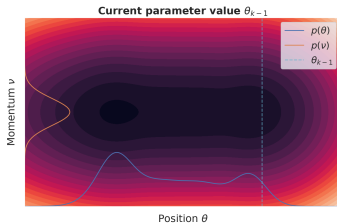
Advanced sampling methods: Hamiltonian Monte Carlo

Hamiltonian dynamics as a proposal distribution



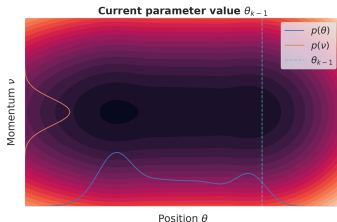
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Hamiltonian dynamics as a proposal distribution



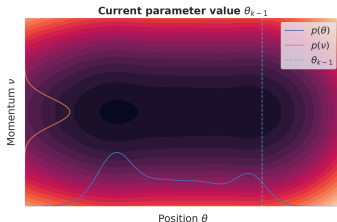
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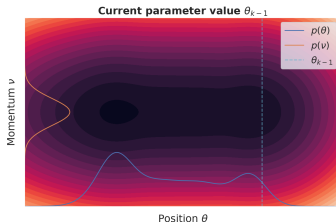
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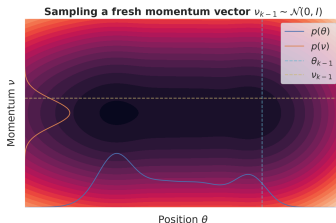
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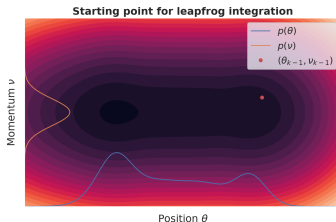
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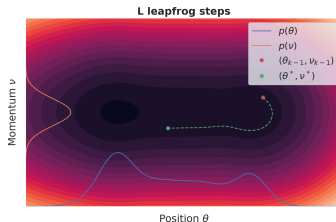
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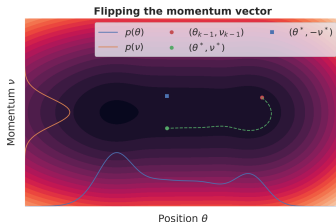
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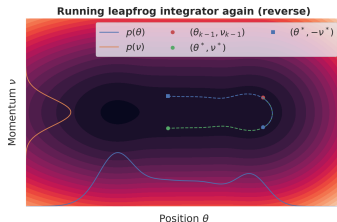
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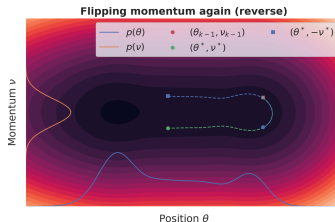
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The big picture of HMC

Hamiltonian Monte Carlo algorithm

Let $\theta_k \in \mathbb{R}^D$ be the position at iteration k and let $\text{leapfrog}_{\eta,L}(\cdot, \cdot)$ be a leapfrog integrator, then each iteration of HMC proceeds as follows

For $k = 1, 2, \dots, K$, do

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The big picture of HMC

Hamiltonian Monte Carlo algorithm

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For $k = 1, 2, \dots, K$, do

1. Sample *new momentum* variable $\nu_{k-1} \sim \mathcal{N}(\mathbf{0}, I)$
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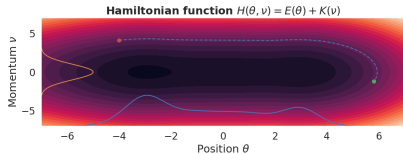
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- Note that if step-size η is sufficiently small such that $H(\theta^*, \nu^*) \approx H(\theta_{k-1}, \nu_{k-1})$, the acceptance probability will be 100%!

Tuning HMC

■ *Hyperparameters* of HMC

- Step size $\eta > 0$ for leap frog integrator
- Number of leapfrog steps L



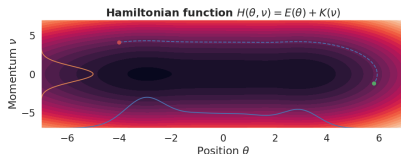
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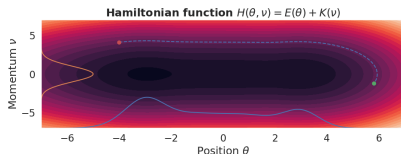
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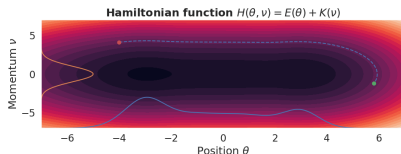
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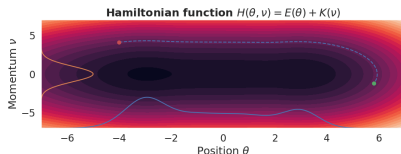
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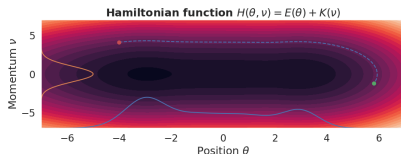
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- A simpler option is either to *fix* $L \in \{10, \dots, 100\}$ or *randomize* L every iteration.

Advanced sampling methods: The MALA and ULA algorithms

The MALA algorithm

- HMC has a little brother: MALA. Recall the leapfrog equations

$$\boldsymbol{\nu}_{k+1/2} = \boldsymbol{\nu}_k - \frac{\eta}{2} \nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}_k) \quad (\text{Half-step for } \boldsymbol{\nu})$$

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- Like MRW, but uses the gradient to bias the proposals towards high density regions

MALA and ULA

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- ULA is therefore faster, but it is also *biased* in the sense that it will not converge to the true target distribution

Overview

Algorithm	Proposal	MH Step?	Properties
MRW	$\theta^* = \theta_k + \eta \epsilon$ $\epsilon \sim \mathcal{N}(0, I)$	Yes	Random walk. Can suffer from slow exploration.
ULA	$\theta_{k+1} = \theta_k - \frac{\eta^2}{2} \nabla E(\theta_k) + \eta \epsilon$ $\epsilon \sim \mathcal{N}(0, I)$	No	Unadjusted. Follows the gradient for directed exploration. Always accepts, resulting in a <i>biased</i> stationary distribution.
MALA	$\theta^* = \theta_k - \frac{\eta^2}{2} \nabla E(\theta_k) + \eta \epsilon$ $\epsilon \sim \mathcal{N}(0, I)$	Yes	Adjusted with MH correction. Eliminates ULA's bias. Equivalent to HMC with $L = 1$. Scales better than MRW.
HMC	$(\theta^*, \nu^*) = \text{Leapfrog}_{\eta, L}(\theta_k, \nu_k)$ $\nu_k \sim \mathcal{N}(0, I)$	Yes	Simulates Hamiltonian dynamics. Suppresses random walks, allowing for distant proposals with high acceptance rates.

Convergence diagnostics

Are we there yet?

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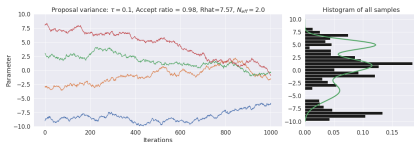
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- Let B denote the *between-chain variance* and let W denote the *within-chain variance*, then $\hat{R}$ -statistic (or the *potential scale reduction factor*) for chains of length N is defined by

$$\hat{R}^2 = \frac{S-1}{S} + \frac{1}{S} \frac{B}{W}$$



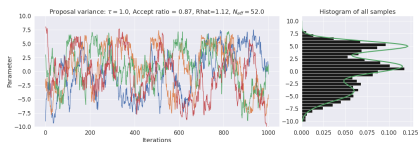
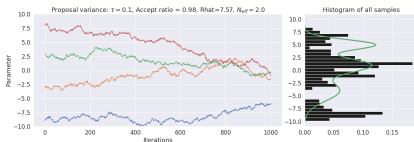
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- If $B = W$, then $\hat{R} = 1$. If $B > W$, then $\hat{R} > 1$. In practice, we say that the *chains have mixed* if $\hat{R} < 1.1$



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How accurate is MCMC? Quantifying the error

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Example: MCMC diagnostics

- The $\hat{R}$ - statistic

$$\hat{R}^2 = \frac{N-1}{N} + \frac{1}{N} \frac{B}{W}$$

- The effective sample size

$$S_{\text{eff}} = \frac{S}{1 + 2 \sum_{t=1}^{\infty} \rho_t}$$

- The Monte Carlo Standard Error

$$\text{MCSE} = \frac{1}{\sqrt{S_{\text{eff}}}} \widehat{\text{sd}}(f(w))$$



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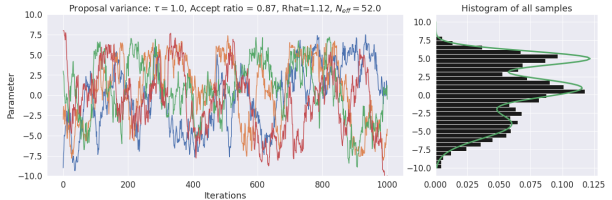
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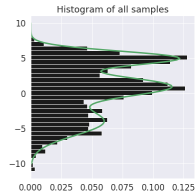
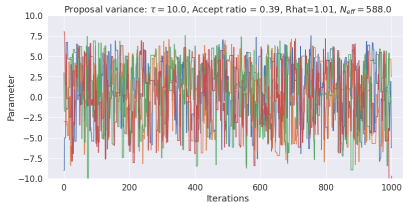
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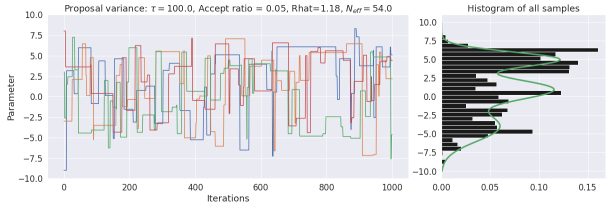
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- The transition kernel for Metropolis-Hasting

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The detailed balance condition III

- We just arrived at

$$p(\theta^k)T(\theta^*|\theta^k) = p(\theta^*)q(\theta^k|\theta^*)$$

- What about the opposite direction?

$$T(\theta^k|\theta^*) = q(\theta^k|\theta^*)A(\theta^k|\theta^*) = q(\theta^k|\theta^*)$$

The detailed balance condition III

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$$p(\theta^k) T(\theta^* | \theta^k) = p(\theta^*) q(\theta^k | \theta^*)$$

- What about the opposite direction?

$$T(\theta^k | \theta^*) = q(\theta^k | \theta^*) A(\theta^k | \theta^*) = q(\theta^k | \theta^*)$$

- Combining the two and we are done

$$p(\theta^k) T(\theta^* | \theta^k) = p(\theta^*) T(\theta^k | \theta^*)$$